Experiment 4 - Counting the frequency of occurrence of a word in a body of text is often needed during text processing.

Aim:

To count the frequency of occurrence of a word in a body of text.

Program:

```
import nltk
from nltk. tokenize import word_tokenize
from nltk. corpus import gutenberg

nltk.download('gutenberg')
nltk.download('punkt')
nltk.download('punkt_tab')

sample = gutenberg.raw("austen-emma.txt")
token = word_tokenize(sample)
wlist = []

for i in range(50):
        wlist.append(token[i])
wordfreq = [wlist.count(w) for w in wlist]
print("Pairs\n" + str(list(zip(wlist, wordfreq))))
```

Output:

```
Pairs:
('[', 1)          ('Emma', 2)
('by', 1)         ('Jane', 1)
('Austen', 1)     ('1816', 1)
(']', 1)          ('VOLUME', 1)
('I', 2)          ('CHAPTER', 1)
('I', 2)          ('Emma', 2)
('Woodhouse', 1)          (',', 5)
('handsome', 1) (',', 5)
('clever', 1)     (',', 5)
('and', 3)        ('rich', 1)
(',', 5)          ('with', 2)
('a', 1)          ('comfortable', 1)
('home', 1)       ('and', 3)
('happy', 1)      ('disposition', 1)
(',', 5)          ('seemed', 1)
('to', 1)         ('unite', 1)
('some', 1)       ('of', 2)
('the', 2)        ('best', 1)
('blessings', 1)          ('of', 2)
('existence', 1)          (';', 1)
('and', 3)        ('had', 1)
('lived', 1)      ('nearly', 1)
('twenty-one', 1)         ('years', 1)
('in', 1)         ('the', 2)
('world', 1)      ('with', 2)
```

Result:

Thus, we have counted the frequency of occurrence of a word in a body of text.